

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

OCTOBER EXAMINATION

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS

TUESDAY, 15 OCTOBER 2024

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN SOFTWARE ENGINEERING

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMCS2034 INTRODUCTION TO DATA STRUCTURE AND ALGORITHMS**Question 1**

- a) As a software developer working for a ride-sharing company that operates in multiple cities, you are assigned to develop an application that connects drivers with passengers, allowing them to request and provide rides in real time.
- (i) As the applications are getting complex and data-rich, they face significant challenges. Identify and explain **THREE (3)** main challenges related to the application. (9 marks)
 - (ii) Explain and relate **TWO (2)** data structure operations that are suitable to the application. (6 marks)
- b) An inventory management system for a house has been developed by Tess Sdn Bhd to help the inventory management. The system needs to handle different types of products stored in various sections, each with multiple shelves. Your task is to implement *iterators* to facilitate an efficient inventory management system.
- (i) Examine how the iterators can be implemented in the inventory management system. (2 marks)
 - (ii) Assume the inventory is stored in a list of product objects. Write **TWO (2)** methods for an iterator class that iterates over a collection of products in an inventory management system. (4 marks)
 - (iii) *Dynamic programming* and *divide and conquer* are two important paradigms in algorithm design that can be implemented in the inventory management system. Differentiate these paradigms. (4 marks)

[Total: 25 marks]

AMCS2034 INTRODUCTION TO DATA STRUCTURE AND ALGORITHMS**Question 2**

- a) Imagine you are developing a new e-commerce platform that needs to handle large volumes of product data for efficient search and retrieval, which involves an algorithm. The platform allows users to browse and filter products based on various criteria such as price range, category, and availability.
- (i) Examine and explain **TWO (2)** main concerns in the analysis of the algorithm. (6 marks)
 - (ii) Relate the running time analysis with the algorithm selection. (3 marks)
 - (iii) Point out **TWO (2)** problems with the running time analysis for the algorithms of the e-commerce. (4 marks)
 - (iv) Examine how the algorithm's performance under **THREE (3)** different scenarios with different types of input. (9 marks)
- b) Linear search and binary search are used to search a target element within a given dataset typically an array or list. Distinguish both algorithms. (3 marks)

[Total: 25 marks]

Question 3

- a) A set is an efficient data structure to store and process non-duplicate elements. Provide **THREE (3)** concrete classes to create a set. (3 marks)
- b) Using Java programming language, develop a complete program and includes the requirements below:
- Create a *HashSet* to store a list of countries.
 - Add three countries to the *Hashset* known as "New York", "Paris", and "Beijing".
 - Display all the countries stored in the *HashSet* using a loop. (15 marks)
- c) *HashSet* and *TreeSet* are both implementations of the Set interfaces in Java, but they differ in terms of order. Differentiate between *HashSet* and *TreeSet* based on their ordering characteristics. (3 marks)
- d) Data structure is a way of storing and organising data efficiently such that the required operations on them can be performed efficiently. Explain **TWO (2)** types of data structure. (4 marks)

[Total: 25 marks]

Question 4

- a) A graph is a mathematical structure that represents relationships among entities in the real world. A graph consists of a set of edges that connect the vertices.

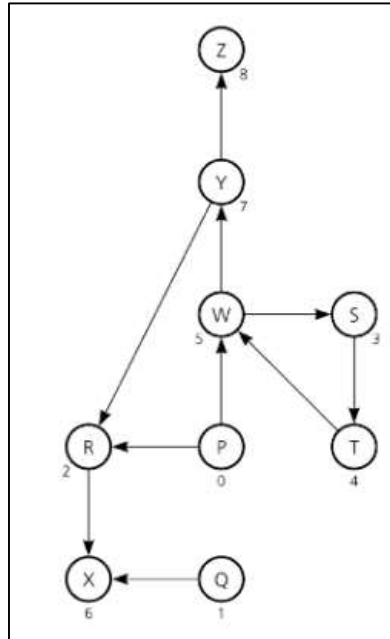


Figure 1

- (i) Develop an adjacency list based on the graph in Figure 1 above. (9 marks)
 - (ii) Develop an adjacency matrix based on the graph in Figure 1 above. (9 marks)
 - (iii) Create an array using the Java program to store the list of vertices based on the graph in Figure 1 above. (3 marks)
- b) Classify and explain the graph (a) and (b) in Figure 2 below as *Depth-first search* or *Breadth-first search*.

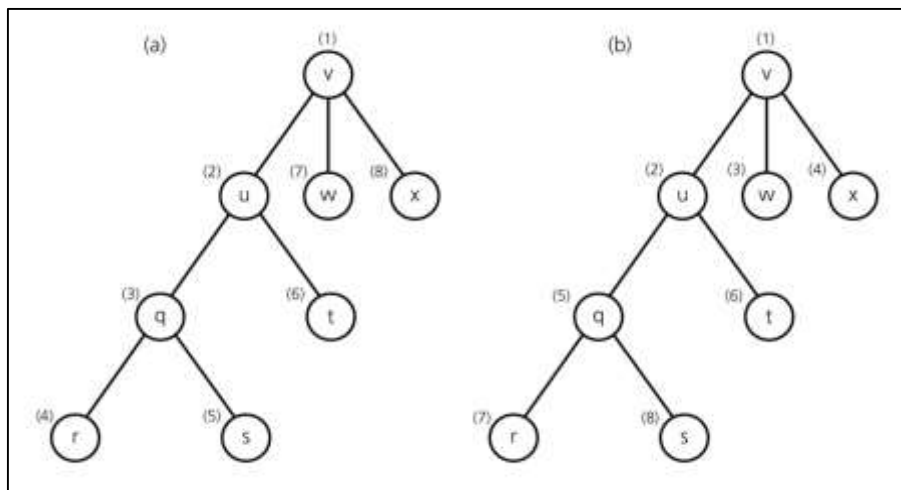


Figure 2

(4 marks)

[Total: 25 marks]